

# 전국기능경기대회 채점기준

## 1. 채점 시 유의사항

직 종 명

클라우드컴퓨팅

※ 다음 사항을 유의하여 채점하십시오.

- 1) AWS의 지역은 모듈에 맞춰 사용합니다.
  - 2) 웹페이지 접근은 크롬이나 파이어폭스를 이용합니다.
  - 3) 웹페이지에서 언어에 따라 문구가 다르게 보일 수 있습니다.
  - 4) shell에서의 명령어의 출력은 버전에 따라 조금 다를 수 있습니다.
  - 6) 채점은 문항 순서대로 진행해야 합니다.
  - 9) 부분 점수가 따로 없는 문항은 모두 맞아야 점수로 인정됩니다.
  - 10) 리소스의 정보를 읽어오는 채점항목은 기본적으로 스크립트 결과를 통해 채점을 진행하며, 만약 선수가 이의가 있다면 명령어를 직접 입력하여 확인해볼 수 있습니다.
  - 11) Cloudshell에서 모듈별 mark.sh를 실행합니다.
  - 12) 채점 시에는 별도로 제공한 모듈별 채점 스크립트(mark.sh)를 실행하여 채점할 수 있습니다.
- 다만, 선수가 직접 입력을 원할 경우 채점기준표에 명시된 명령어 그대로 입력하여 채점할 수 있습니다.

※ 채점기준 양식은 반드시, 양식에 맞추어 작성해야 합니다.(채점사이트 입력에 필요)

2. 채점기준표

| 1) 주요항목별 배점 |          |                   | 직 종 명 |      | 클라우드컴퓨팅 |           |           |    |
|-------------|----------|-------------------|-------|------|---------|-----------|-----------|----|
| 과제<br>번호    | 일련<br>번호 | 주요항목              | 배점    | 채점방법 |         | 채점시기      |           | 비고 |
|             |          |                   |       | 독립   | 합의      | 경기<br>진행중 | 경기<br>종료후 |    |
| 제2과제        | 1        | CDN               | 7.5   | ○    |         |           | ○         |    |
|             | 2        | Keycloak          | 7.5   | ○    |         |           | ○         |    |
|             | 3        | Container Logging | 7.5   | ○    |         |           | ○         |    |
|             | 4        | Workflow          | 7.5   | ○    |         |           | ○         |    |
| 합 계         |          |                   | 30    |      |         |           |           |    |

## 2) 채점방법 및 기준

| 과제<br>번호 | 일<br>련<br>번<br>호 | 주요항목              | 일련<br>번호 | 세부항목(채점방법)                     | 배점  |
|----------|------------------|-------------------|----------|--------------------------------|-----|
| 제1과제     | 1                | CDN               | 1        | S3 Bucket                      | 0.5 |
|          |                  |                   | 2        | CloudFront Distribution        | 1.0 |
|          |                  |                   | 3        | CloudFront Functions           | 1.5 |
|          |                  |                   | 4        | CloudFront Functions Associate | 1.5 |
|          |                  |                   | 5        | Lambda@edge                    | 1.5 |
|          |                  |                   | 6        | Resizing                       | 1.5 |
|          | 2                | Keycloak          | 1        | VPC                            | 0.5 |
|          |                  |                   | 2        | Keycloak Server                | 1.0 |
|          |                  |                   | 3        | Keycloak ALB                   | 1.5 |
|          |                  |                   | 4        | Keycloak Setting               | 1.5 |
|          |                  |                   | 5        | AWS IAM                        | 1.5 |
|          |                  |                   | 6        | Keycloak SAML                  | 1.5 |
|          | 3                | Container Logging | 1        | Application Running            | 0.5 |
|          |                  |                   | 2        | Fluentbit Pod                  | 1.0 |
|          |                  |                   | 3        | Otel Pod                       | 1.0 |
|          |                  |                   | 4        | Loki Pod                       | 1.0 |
|          |                  |                   | 5        | Prometheus Pod                 | 1.0 |
|          |                  |                   | 6        | Datasource                     | 1.5 |
|          |                  |                   | 7        | Grafana dashboard              | 1.5 |
|          | 4                | Workflow          | 1        | S3 Bucket                      | 0.5 |
|          |                  |                   | 2        | DynamoDB                       | 1.0 |
|          |                  |                   | 3        | Lmabda Functions               | 1.5 |
|          |                  |                   | 4        | Lambda Invoke                  | 1.5 |
|          |                  |                   | 5        | State Machine                  | 1.5 |
|          |                  |                   | 6        | Step Function E2E              | 1.5 |
| 합계       |                  |                   |          |                                | 30  |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1-1 | <pre>1) CN=\$(aws s3api list-buckets --query "Buckets[?contains(Name,'wsc2026-cdn-asset')].Name [0]" --output text   sed 's/wsc2026-cdn-asset-//'); B="wsc2026-cdn-asset-\$CN" echo \$B aws s3api get-bucket-versioning --bucket \$B --query Status --output text aws s3api get-public-access-block --bucket \$B --query '[PublicAccessBlockConfiguration.BlockPublicAcls,PublicAccessBlockConfiguration.BlockPublicPolicy,PublicAccessBlockConfiguration.IgnorePublicAcls,PublicAccessBlockConfiguration.RestrictPublicBuckets]' --output text aws s3 ls s3://\$B/origin/ 2&gt;/dev/null   awk '\$3&gt;0{print \$4}' 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-cdn-asset-&lt;선수 비번호&gt; Enabled True    True    True    True worldskills_banner.png</pre> |
| 1-2 | <pre>1) CF_ID=\$(aws resourcegroupstaggingapi get-resources --tag-filters Key=Name,Values=wsc2026-cdn --resource-type-filters cloudfront --query "ResourceTagMappingList[0].ResourceARN" --output text   sed 's:.*://:'); CF_DOMAIN=\$(aws cloudfront get-distribution --id \$CF_ID --query "Distribution.DomainName" --output text) echo "\$CF_DOMAIN"; aws cloudfront get-distribution --id \$CF_ID --query 'Distribution.DistributionConfig.Origins.Items[0].DomainName' --output text; aws cloudfront get-distribution --id \$CF_ID --query 'Distribution.DistributionConfig.PriceClass' --output text 2) 실행결과가 아래와 동일한지 확인합니다. d3j6adxcjmy2xb.cloudfront.net wsc2026-cdn-asset-&lt;비번호&gt;.s3.us-east-1.amazonaws.com PriceClass_All</pre>    |
| 1-3 | <pre>1) aws cloudfront list-functions --query "FunctionList.Items[?contains(Name,'wsc2026')].{Name:Name,Stage:FunctionMetadata.Stage}" --output text   grep LIVE 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-device-detect    LIVE wsc2026-response-header LIVE</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1-4 | <pre>1) aws cloudfront get-distribution --id \$CF_ID --query "Distribution.DistributionConfig.DefaultCacheBehavior.FunctionAssociations.Items[].[EventType,FunctionARN]" --output text 2) 실행 결과가 아래와 동일한지 확인합니다. viewer-request  arn:aws:cloudfront::&lt;ACCOUNT_ID&gt;:function/wsc2026-device-detect viewer-response arn:aws:cloudfront::&lt;ACCOUNT_ID&gt;:function/wsc2026-response-header</pre>                                                                                                                                                                                                                                                                                                                                               |
| 1-5 | <pre>1) aws lambda get-function --function-name wsc2026-resize --query "Configuration.[FunctionName,Runtime]" --output text 2&gt;/dev/null RN=\$(aws lambda get-function --function-name wsc2026-resize --query "Configuration.Role" --output text 2&gt;/dev/null   awk -F/ '{print \$NF}'); echo "\$RN"; aws iam list-attached-role-policies --role-name "\$RN" --query 'AttachedPolicies[.PolicyName]' --output text 2&gt;/dev/null aws cloudfront get-distribution --id \$CF_ID --query "Distribution.DistributionConfig.DefaultCacheBehavior.LambdaFunctionAssociations.Items[].[EventType,LambdaFunctionARN]" --output</pre>                                                                                                                  |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p>text 2&gt;/dev/null</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <pre>wsc2026-resize python3.12 wsc2026-resize-role wsc2026-resize-policy origin-response arn:aws:lambda:us-east-1:&lt;ACCOUNT_ID&gt;:function:wsc2026-resize:&lt;Version&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1-6 | <p>1) 0F="origin/\$(aws s3 ls s3://\$B/origin/ 2&gt;/dev/null   awk '\$3&gt;0{print \$4}'   head -1)";<br/> echo "\$0F \$(aws s3api head-object --bucket \$B --key "\$0F" --query ContentLength<br/> --output text 2&gt;/dev/null)"</p> <pre>NOW_UTC=\$(date -u +%s); NOW_KST=\$(date +%s); ME=113209; DE=121900; BUST=\$(date +%s) curl -s -o /tmp/m.png -D /tmp/m.txt -A "Mozilla/5.0 (iPhone; CPU iPhone OS 17_0 like Mac OS X)" "https://\$CF_DOMAIN/\$0F?w=480&amp;h=320&amp;type=mobile&amp;bust=\$BUST"; echo "mobile \$(wc -c&lt;/tmp/m.png) \$(grep -i x-device-type /tmp/m.txt 2&gt;/dev/null awk '{print \$2}' tr -d '\r') \$(grep -i x-resized /tmp/m.txt 2&gt;/dev/null awk '{print \$2}' tr -d '\r')"</pre> <pre>curl -s -o /tmp/d.png -D /tmp/d.txt -A "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 Chrome/120.0" "https://\$CF_DOMAIN/\$0F?w=1920&amp;h=1080&amp;type=desktop&amp;bust=\$BUST"; echo "desktop \$(wc -c&lt;/tmp/d.png) \$(grep -i x-device-type /tmp/d.txt 2&gt;/dev/null awk '{print \$2}' tr -d '\r') \$(grep -i x-resized /tmp/d.txt 2&gt;/dev/null awk '{print \$2}' tr -d '\r')"</pre> <pre>aws s3 ls s3://\$B/resized/ 2&gt;/dev/null   awk '\$3&gt;0{print \$1,\$2,\$4}'   tail -2   while read D T F; do SD=\$(( NOW_UTC - \$(date -u -d "\$D \$T" +%s) )); TS=\$(echo \$F grep -oP '\d{8}_\d{6}' sed 's/\(...\)\(...\)\(...\)\(...\)\(...\)\(...\)/\1-\2-\3 \4:\5:\6/'); FD=\$(( NOW_KST - \$(date -d "\$TS" +%s) )); SZ=\$(aws s3api head-object --bucket \$B --key "resized/\$F" --query ContentLength --output text 2&gt;/dev/null); EX=\$(echo \$F grep -q "^mobile" &amp;&amp; echo \$ME    echo \$DE); [ "\$SD" -le 60 ] &amp;&amp; [ "\$FD" -le 60 ] &amp;&amp; [ "\$SZ" -eq "\$EX" ] &amp;&amp; echo "\$F : PASS"    echo "\$F : FAIL (s3=\${SD}s fname=\${FD}s size=\${SZ} expected=\${EX})"; done</pre> <p>2) 실행결과가 아래와 동일한지 확인합니다. S3 객체명은 KST 기준으로 현재 날짜와 같고, 현재<br/> 시간 5분 내외인지 확인합니다.</p> <pre>origin/worldskills_banner.png 111811 mobile 113209 mobile true desktop 121900 desktop true desktop_worldskills_banner_20260530_113354.png : PASS mobile_worldskills_banner_20260530_113351.png : PASS</pre> |
| 2-1 | <p>1) VID=\$(aws ec2 describe-vpcs --filters Name=tag:Name,Values=wsc2026-keycloak-vpc --query<br/> 'Vpcs[0].VpcId' --output text --region ap-northeast-2)<br/> aws ec2 describe-vpcs --vpc-ids \$VID --query 'Vpcs[0].[CidrBlock,Tags[?Key=`Name`]][0].Va<br/> lue]' --output text --region ap-northeast-2<br/> aws ec2 describe-subnets --filters Name=vpc-id,Values=\$VID --query 'Subnets[].[Tags[?Key=<br/> =`Name`]][0].Value,CidrBlock,AvailabilityZone]' --output text --region ap-northeast-2   so<br/> rt</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

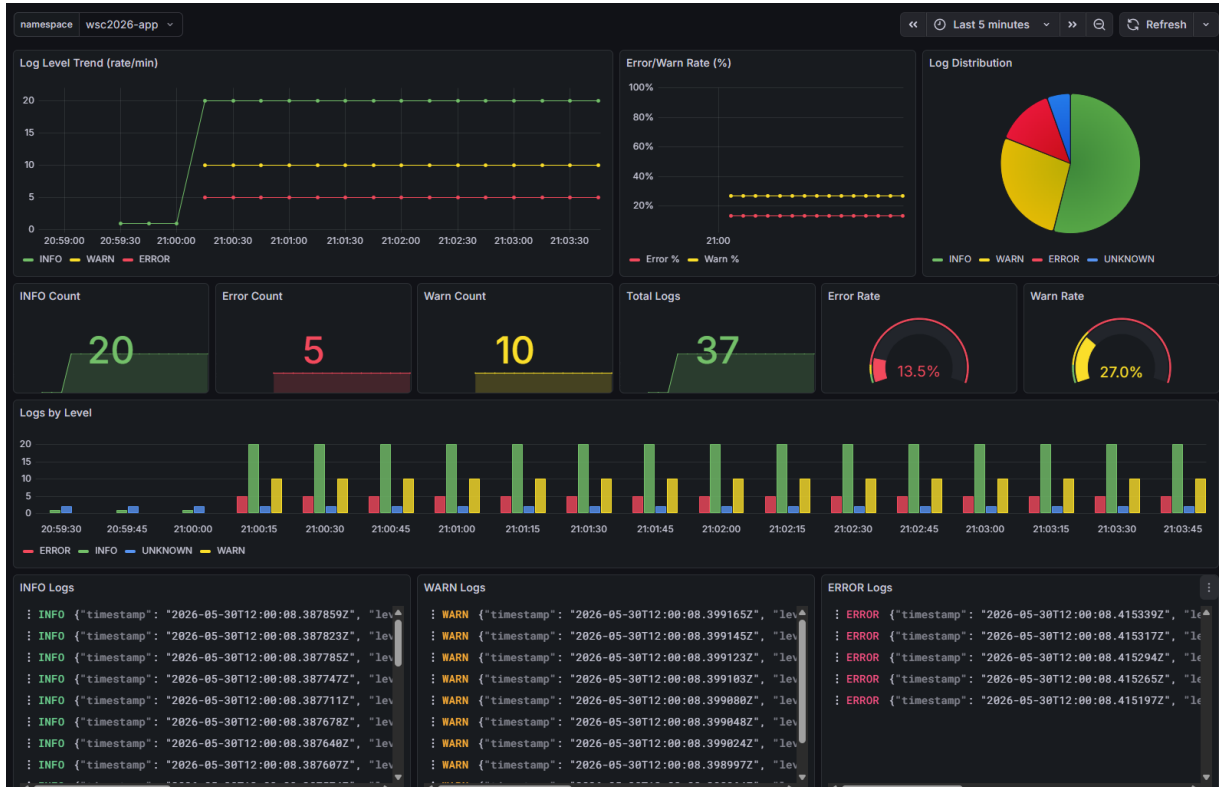
| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | 10.20.0.0/16      wsc2026-keycloak-vpc<br>wsc2026-private-subnet-a      10.20.10.0/24      ap-northeast-2a<br>wsc2026-public-subnet-a 10.20.1.0/24      ap-northeast-2a<br>wsc2026-public-subnet-b 10.20.2.0/24      ap-northeast-2b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2-2 | <pre> 1) IID=\$(aws ec2 describe-instances --filters Name=tag:Name,Values=wsc2026-keycloak Name=i nstance-state-name,Values=running --query 'Reservations[0].Instances[0].InstanceId' --outp ut text --region ap-northeast-2) aws ec2 describe-instances --instance-ids \$IID --query 'Reservations[0].Instances[0].[Tags [?Key=`Name`] [0].Value,InstanceType,State.Name]' --output text --region ap-northeast-2 SID=\$(aws ec2 describe-instances --instance-ids \$IID --query 'Reservations[0].Instances [0].SubnetId' --output text --region ap-northeast-2) echo "Subnet: \$(aws ec2 describe-subnets --subnet-ids \$SID --query 'Subnets[0].Tags[?Key= =`Name`] [0].Value' --output text --region ap-northeast-2)" EC2SG=\$(aws ec2 describe-instances --instance-ids \$IID --query 'Reservations[0].Instances [0].SecurityGroups[0].GroupId' --output text --region ap-northeast-2) echo "EC2-SG: \$([ "\$(aws ec2 describe-security-groups --group-ids \$EC2SG --query 'Security Groups[0].GroupName' --output text --region ap-northeast-2)" = wsc2026-keycloak-sg ] &amp;&amp; ec ho PASS    echo FAIL)" 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-keycloak      t3.medium      running Subnet: wsc2026-private-subnet-a EC2-SG: PASS </pre>                                                                                     |
| 2-3 | <pre> 1) ALB=\$(aws elbv2 describe-load-balancers --names wsc2026-keycloak-alb --query 'LoadBalan cers[0]' --output json --region ap-northeast-2) echo "\$ALB"   python3 -c "import sys,json;d=json.load(sys.stdin);print(d['LoadBalancerName ']+'\t'+d['Scheme'])" ALBARN=\$(echo "\$ALB"   python3 -c "import sys,json;print(json.load(sys.stdin)['LoadBalance rArn'])") TG=\$(aws elbv2 describe-target-groups --names wsc2026-keycloak-tg --query 'TargetGroups [0].TargetGroupArn' --output text --region ap-northeast-2) echo "Listener: \$(aws elbv2 describe-listeners --load-balancer-arn \$ALBARN --query 'Listen ers[?Port=`80`].Port' --output text --region ap-northeast-2)" echo "TargetGroup: wsc2026-keycloak-tg" echo "TargetHealth: \$(aws elbv2 describe-target-health --target-group-arn \$TG --query 'Tar getHealthDescriptions[0].TargetHealth.State' --output text --region ap-northeast-2)" ALBSG=\$(echo "\$ALB"   python3 -c "import sys,json;print(json.load(sys.stdin)['SecurityGrou ps'][0])") echo "ALB-SG: \$([ "\$(aws ec2 describe-security-groups --group-ids \$ALBSG --query 'Security Groups[0].GroupName' --output text --region ap-northeast-2)" = wsc2026-keycloak-alb-sg ] &amp; &amp; echo PASS    echo FAIL)" 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-keycloak-alb      internet-facing </pre> |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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|     | <p>Listener: 80</p> <p>TargetGroup: wsc2026-keycloak-tg</p> <p>TargetHealth: healthy</p> <p>ALB-SG: PASS</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2-4 | <pre> 1) URL=http://\$(aws elbv2 describe-load-balancers --names wsc2026-keycloak-alb --query 'LoadBalancers[0].DNSName' --output text --region ap-northeast-2) TOKEN=\$(curl -s -X POST "\$URL/realms/master/protocol/openid-connect/token" -d "client_id=admin-cli&amp;grant_type=password&amp;username=admin&amp;password=Skill53#!@#"   python3 -c "import sys,json;print(json.load(sys.stdin).get('access_token',''))") echo "wsc2026-aws Realm: \$(curl -s -o /dev/null -w '%{http_code}' -H "Authorization: Bearer \$TOKEN" \$URL/admin/realms/wsc2026-aws)" echo "Groups:" curl -s -H "Authorization: Bearer \$TOKEN" "\$URL/admin/realms/wsc2026-aws/groups"   python3 -c "import sys,json;[print(' '+g['name']) for g in json.load(sys.stdin)]" echo "Users:" for U in dev-user infra-user; do     echo " \$U: \$(curl -s -o /dev/null -w '%{http_code}' -H "Authorization: Bearer \$TOKEN" "\$URL/admin/realms/wsc2026-aws/users?username=\$U&amp;exact=true")" done echo "Login:" echo " dev-user: \$(curl -s -o /dev/null -w '%{http_code}' -X POST "\$URL/realms/wsc2026-aws/protocol/openid-connect/token" -d "client_id=admin-cli&amp;grant_type=password&amp;username=dev-user&amp;password=Skills_dev53%25%24%25" )" echo " infra-user: \$(curl -s -o /dev/null -w '%{http_code}' -X POST "\$URL/realms/wsc2026-aws/protocol/openid-connect/token" -d "client_id=admin-cli&amp;grant_type=password&amp;username=infra-user&amp;password=Skills_infra53%23%40 %23")" 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-aws Realm: 200 Groups:     dev-team     infra-team Users:     dev-user: 200     infra-user: 200 Login:     dev-user: 200     infra-user: 200 </pre> |
| 2-5 | <pre> 1) echo "SAML: \$(aws iam list-saml-providers --query 'SAMLProviderList[0].Arn' --output te xt   awk -F/ '{print \$NF}')" echo "Roles:" </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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|     | <pre>aws iam get-role --role-name wsc2026-dev-role --query 'Role.RoleName' --output text   sed 's/^/ /'</pre> <pre>aws iam get-role --role-name wsc2026-infra-role --query 'Role.RoleName' --output text   sed 's/^/ /'</pre> <pre>echo "Policies:"</pre> <pre>aws iam list-policies --scope Local --query "Policies[?PolicyName=='wsc2026-dev-policy'].PolicyName" --output text   sed 's/^/ /'</pre> <pre>aws iam list-policies --scope Local --query "Policies[?PolicyName=='wsc2026-infra-policy'].PolicyName" --output text   sed 's/^/ /'</pre> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>SAML: wsc2026-keycloak-idp</p> <p>Roles:</p> <ul style="list-style-type: none"> <li>wsc2026-dev-role</li> <li>wsc2026-infra-role</li> </ul> <p>Policies:</p> <ul style="list-style-type: none"> <li>wsc2026-dev-policy</li> <li>wsc2026-infra-policy</li> </ul>                                                                                                                                                                                                                                                                                            |
| 2-6 | <p>1) <code>aws ec2 run-instances --image-id \$(aws ssm get-parameter --name /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 --query Parameter.Value --output text --region ap-northeast-2) --instance-type t3.micro --subnet-id \$(aws ec2 describe-subnets --filters Name=tag:Name,Values=wsc2026-private-subnet-a --query 'Subnets[0].SubnetId' --output text) --tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=wsc2026-protected-ec2},{Key=protected,Value=true}]' --region ap-northeast-2 --no-cli-pager &gt; /dev/null 2&gt;&amp;1</code><br/> <code>echo "[수동] Login URL: \$URL/realms/wsc2026-aws/protocol/saml/clients/amazon-aws"</code></p> <p>2) 출력된 URL을 통해 infra-user로 로그인하여 AWS 콘솔에 로그인하여 계정이 wsc2026-infra-role/infra-user인지 확인합니다. 확인이 되었다면 서울 외 리전에 인스턴스를 조회하여 성공적으로 조회되는지 확인합니다. 그리고 서울 리전으로 돌아와 wsc2026-protected-ec2 인스턴스를 중지하고 오류가 발생하며 중지가 되지 않는지 확인합니다.</p> <p>3) 출력된 URL을 통해 dev-user로 로그인하여 AWS 콘솔에 로그인하여 계정이 wsc2026-dev-role/dev-user인지 확인합니다. 확인이 되었다면 서울 리전에 인스턴스를 성공적으로 조회하지만, 도쿄 리전에 인스턴스는 조회되지 않는지 확인합니다. 마찬가지로 서울 리전에 s3 버킷이 조회가 되어야하지만, 도쿄 리전에선 s3 버킷이 조회되지 않는지 확인합니다.</p> |
| 3-1 | <p>1) <code>C='wsc2026-logging-cluster'; N='wsc2026-logging'; OC='wsc2026-otel-collector-opentelemetry-collector'</code><br/> <code>aws eks update-kubeconfig --name \$C --region ap-northeast-1 2&gt;/dev/null</code><br/> <code>kubectl get deploy -n wsc2026-app --no-headers 2&gt;/dev/null   awk '{print \$1, \$2}'</code><br/> <code>LB=\$(kubectl get svc log-generator -n wsc2026-app -o jsonpath='{.status.loadBalancer.ingress[0].hostname}' 2&gt;/dev/null)</code><br/> <code>curl -s "http://\$LB/health" 2&gt;/dev/null; echo</code></p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <pre>log-generator 1/1 {"status": "ok"}</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 3-2 | <p>1) echo -n "FluentBit: "; kubectl get pods -n \$N -l app.kubernetes.io/name=fluent-bit --no-headers 2&gt;/dev/null   grep -c "Running"</p> <p>echo -n "Nodes: "; kubectl get nodes --no-headers 2&gt;/dev/null   wc -l</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>FluentBit: 2</p> <p>Nodes: 2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3-3 | <p>1) echo -n "OTel Collector: "; kubectl get pods -n \$N -l app.kubernetes.io/name=opentelemetry-collector --no-headers 2&gt;/dev/null   grep -c "Running"</p> <p>kubectl logs -n \$N deploy/\$OC --tail=50 2&gt;/dev/null   grep -o 'log records': [0-9]*   tail -1</p> <p>2) 실행 결과에서 Otel Collector와 log records에서 1 이상이 출력되는지 확인합니다.</p> <p>OTel Collector: 1</p> <p>log records": 35</p>                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3-4 | <p>1) echo -n "Loki: "; kubectl get pods -n \$N -l app.kubernetes.io/name=loki --no-headers 2&gt;/dev/null   grep -c "Running"</p> <p>echo -n "Labels: "; kubectl exec deploy/wsc2026-prometheus-server -n \$N -c prometheus-server -- wget -qO- "http://wsc2026-loki.\$N.svc.cluster.local:3100/loki/api/v1/labels" 2&gt;/dev/null   python3 -c "import sys,json;print(len(json.load(sys.stdin).get('data',[])))" 2&gt;/dev/null</p> <p>2) 실행 결과에서 Loki와 Labels에서 1 이상이 출력되는지 확인합니다.</p> <p>Loki: 4</p> <p>Labels: 4</p>                                                                                                                                                                                                                                                                         |
| 3-5 | <p>1) curl -s "http://\$LB/burst?count=20&amp;level=INFO" &gt;/dev/null ; curl -s "http://\$LB/burst?count=10&amp;level=WARN" &gt;/dev/null ; curl -s "http://\$LB/burst?count=5&amp;level=ERROR" &gt;/dev/null sleep 15</p> <p>echo -n "Prometheus: "; kubectl get pods -n \$N -l app.kubernetes.io/name=prometheus --no-headers 2&gt;/dev/null   grep -c "Running"</p> <p>kubectl exec deploy/wsc2026-prometheus-server -n \$N -c prometheus-server -- \</p> <p>wget -qO- "http://localhost:9090/api/v1/query?query=log_record_count_total" \</p> <p>2&gt;/dev/null   python3 -c "import sys,json;r=json.load(sys.stdin);print('Metrics: '+str(len(r['data']['result'])))" 2&gt;/dev/null</p> <p>2) 실행 결과에서 Prometheus와 Metrics에서 1 이상이 출력되는지 확인합니다.</p> <p>Prometheus: 1</p> <p>Metrics: 3</p> |
| 3-6 | <p>1) GF=\$(kubectl get svc wsc2026-grafana -n \$N -o jsonpath='{.status.loadBalancer.ingress[0].hostname}' 2&gt;/dev/null)</p> <p>echo "Datasources:"</p> <p>curl -s -u admin:Skill53@@ "http://\$GF/api/datasources" 2&gt;/dev/null   python3 -c "import sys,json;[print(' '+d['name']+' ('+d['type']+')') for d in json.load(sys.stdin)]" 2&gt;/dev/null</p> <p>echo "Dashboards:"</p> <p>curl -s -u admin:Skill53@@ "http://\$GF/api/search?query=wsc2026" 2&gt;/dev/null   python3 -c</p>                                                                                                                                                                                                                                                                                                    |

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|     | <p>"import sys,json;[print(' '+d['title']) for d in json.load(sys.stdin)]" 2&gt;/dev/null</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>Datasources:</p> <p>Loki (loki)</p> <p>Prometheus (prometheus)</p> <p>Dashboards:</p> <p>wsc2026-app-logs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3-7 |  <p>The screenshot displays a Grafana dashboard for the 'wsc2026-app-logs' namespace. It features several panels: 'Log Level Trend (rate/min)' showing a sharp increase in INFO logs at 21:00:00; 'Error/Warn Rate (%)' showing a line graph for error and warn rates; 'Log Distribution' showing a pie chart for INFO, WARN, ERROR, and UNKNOWN log levels; 'INFO Count' (20), 'Error Count' (5), 'Warn Count' (10), and 'Total Logs' (37) as large green numbers; 'Error Rate' (13.5%) and 'Warn Rate' (27.0%) as gauge charts; 'Logs by Level' as a bar chart showing log frequency over time; and three panels at the bottom showing 'INFO Logs', 'WARN Logs', and 'ERROR Logs' with their respective timestamps and levels.</p> |
| 4-1 | <p>1) B=\$(aws s3api list-buckets --query "Buckets[?contains(Name,'wsc2026-order-pipeline')].Name[0]" --output text)</p> <p>echo \$B</p> <p>aws s3api get-bucket-versioning --bucket \$B --query Status --output text 2&gt;/dev/null</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>wsc2026-order-pipeline</p> <p>Enabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4-2 | <p>1) for T in wsc2026-orders wsc2026-inventory wsc2026-pipeline-history; do</p> <p>    K=\$(aws dynamodb describe-table --table-name \$T --query "Table.KeySchema[].[KeyType,AttributeName]" --output text 2&gt;/dev/null)</p> <p>    PK=\$(echo "\$K" grep HASH awk '{print \$2}')</p> <p>    SK=\$(echo "\$K" grep RANGE awk '{print \$2}')</p> <p>    echo "\$T       \$PK       \$SK"</p>                                                                                                                                                                                                                                                                                                                                                                                                                          |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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|     | <p>done</p> <p>aws dynamodb describe-time-to-live --table-name wsc2026-pipeline-history --query 'TimeToLiveDescription.TimeToLiveStatus' --output text 2&gt;/dev/null</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>wsc2026-orders    order_id            ordered_at</p> <p>wsc2026-inventory    product_id</p> <p>wsc2026-pipeline-history    execution_id    started_at</p> <p>ENABLED</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4-3 | <p>1) for F in wsc2026-order-validator wsc2026-payment-processor; do<br/>  echo "\$F    \$(aws lambda get-function --function-name \$F --query 'Configuration.Runtime' --output text 2&gt;/dev/null)"</p> <p>done</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>wsc2026-order-validator python3.13</p> <p>wsc2026-payment-processor    python3.13</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 4-4 | <p>1) aws lambda invoke --function-name wsc2026-order-validator --payload '{"order_id":"ORD-T1","product_id":"P1","quantity":2,"unit_price":1000,"payment_method":"CARD"}' /tmp/v --cli-binary-format raw-in-base64-out &gt;/dev/null 2&gt;&amp;1 &amp;&amp; jq -r "'VALID\t'+(.valid tostring)+'\tERRORS='+(.errors length) tostring)'" /tmp/v</p> <p>aws lambda invoke --function-name wsc2026-order-validator --payload '{"order_id":"BAD","product_id":"","quantity":0,"unit_price":-1,"payment_method":"X"}' /tmp/i --cli-binary-format raw-in-base64-out &gt;/dev/null 2&gt;&amp;1 &amp;&amp; jq -r "'INVALID\t'+(.valid tostring)+'\tERRORS='+(.errors length) tostring)'" /tmp/i</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>VALID    true    ERRORS=0</p> <p>INVALID false   ERRORS=5</p>                     |
| 4-5 | <p>1) S=\$(aws stepfunctions list-state-machines --query "stateMachines[?name=='wsc2026-order-pipeline'].stateMachineArn [0]" --output text)</p> <p>aws stepfunctions describe-state-machine --state-machine-arn \$S --query '[name,type]' --output text 2&gt;/dev/null</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>wsc2026-order-pipeline    STANDARD</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4-6 | <p>1) E=\$(aws stepfunctions start-execution --state-machine-arn \$S --input '{"bucket\":\"\$B\",\"key\":\"incoming/sample-orders.json\"}' --query executionArn --output text 2&gt;/dev/null); sleep 30</p> <p>aws stepfunctions describe-execution --execution-arn \$E --query status --output text 2&gt;/dev/null</p> <p>aws dynamodb scan --table-name wsc2026-orders --query Count --output text 2&gt;/dev/null</p> <p>A=\$(aws dynamodb get-item --table-name wsc2026-inventory --key '{"product_id":{"S":"PROD-A100"}}' --query Item.stock.N --output text 2&gt;/dev/null); B2=\$(aws dynamodb get-item --table-name wsc2026-inventory --key '{"product_id":{"S":"PROD-B200"}}' --query Item.stock.N --output text 2&gt;/dev/null); C=\$(aws dynamodb get-item --table-name wsc2026-inventory --key ' ')</p> |

| 순번 | 채점 항목                                                                                                                                                                                                                                                                                                                                                                         |
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|    | <pre>{"product_id":{"S":"PROD-C300"}}' --query Item.stock.N --output text 2&gt;/dev/null); echo "\$A \$B2 \$C" aws dynamodb scan --table-name wsc2026-pipeline-history --query 'Items[-1].[status.S,total_orders.N,processed_orders.N]' --output text 2&gt;/dev/null</pre> <p>2) 실행결과가 아래와 동일한지 확인합니다.</p> <pre>SUCCEEDED 2 48      19      100 COMPLETED      5      5</pre> |